

Arizona State University
Department of Economics
Ph.D. Qualifying Exam

Macroeconomic Theory

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Instructions: Read the questions carefully, show all your work, and explain all your answers. Answer any THREE of the following FOUR questions. Indicate carefully if you introduce notation, or make assumptions when answering a question.

Good luck!

1. Consider a version of the growth model with N households. The utility function is the same for all households:

$$\sum_{t=0}^{\infty} \beta^t \log(c_{tn}),$$

where $\beta \in (0, 1)$ and c_{tn} is the consumption of household $n \in \{1, \dots, N\}$. Household n is endowed with an initial capital stock $k_{0n} > 0$, which may differ across households. Each household is also endowed with one unit of time in each period.

The aggregate production function is

$$Y_t = K_t^\theta L_t^{1-\theta},$$

where K_t is aggregate capital and L_t aggregate labor. Output can be used for consumption and investment. Capital depreciates at rate δ .

- a. Write down the planner's problem.
- b. Show that Pareto efficient allocations have the property that in each period household n 's consumption is a constant fraction of aggregate consumption.

- c. Show that in a Pareto efficient allocation, sequences of aggregate capital depend on the initial aggregate capital stock, $K_0 \equiv \sum_{n=1}^N k_{0n}$, but not on the initial distribution of K_0 among households, $\{k_{01}, \dots, k_{0N}\}$.
- d. Now let N be sufficiently large so that the effect of household's choices on aggregate variables is negligible. Define an Arrow-Debreu equilibrium. Under what conditions will a competitive-equilibrium sequence of the aggregate capital stock be independent of the initial distribution of K_0 among households, $\{k_{01}, \dots, k_{0N}\}$? ?

2. Consider the following version of the overlapping generations model with capital. Time is discrete ($t = 0, 1, 2, \dots$). Individuals live for two periods, and work in only the first period of their lives. Population grows at the rate n .

Competitive firms have access to a Cobb-Douglas technology that uses capital (K) and labor of two types; U and S , for *unskilled* and *skilled* labor. Formally, the production function is given by

$$F(K, S, U) = K^\alpha L^{1-\alpha}; \quad L \equiv S^\phi U^{1-\phi}.$$

At birth, individuals are born with either skilled or unskilled efficiency units (labor endowments). A skilled individual is born with z_s efficiency units while an unskilled individual is born with z_u efficiency units. In each cohort, there is a fraction π of skilled individuals and a fraction $1 - \pi$ of unskilled individuals. An individual born at date t chooses consumption levels $c_{1,t}$, $c_{2,t+1}$ and savings s_t , to maximize

$$U(c_{1,t}, c_{2,t+1}) = \log(c_{1,t}) + \beta \log(c_{2,t+1}).$$

- a. Formulate the problem of a young individual of each type in period t .
- b. Define a competitive equilibrium.
- c. We concentrate now on the effects on steady states of a change in level of efficiency units of skilled individuals, z_s . Consider two values; $z_s^+ > z_s^-$. How do the steady-state rates of return on capital compare?

Economist A argues that the economy with z_s^+ has a higher *skill premium*, or labor earnings per skilled worker relative to labor earnings per unskilled worker. Is she right? Explain.

ratio is constant

3. Consider the following version of the standard asset-pricing model. There are two types of trees. The first tree yields a non-storable sequence of dividends $\{x_{1,t}\}$. The second tree yields non-storable dividend sequence $\{x_{2,t}\}$. Dividends of each tree take strictly positive and bounded values, and follow a Markov process with transition function $Q_i(\cdot|x_i)$, $i = 1, 2$.

Individuals maximize the expected discounted utility of consumption. The instantaneous utility of individuals is given by $u(c)$, where c denotes consumption. The function u is increasing, continuous, strictly concave and differentiable. At $t = 0$, each person is endowed with one tree of each type.

a. Formulate the recursive problem for an individual in this economy. Obtain the first order conditions that characterize its solution. Be careful in specifying the relevant state variables and individual decision rules.

b. Define a recursive competitive equilibrium. Derive the functional equations that the price functions for both trees must satisfy.

c. Suppose now that dividends of the second tree are constant over time, and equal to 1. That is, $x_{2,t} = 1$ for all t . In addition, assume that $u(c) = \ln(c)$, and that dividends on the first tree are iid over time with two possible realizations, x_1^H and x_1^L , satisfying $x_1^H > x_1^L$.

Calculate explicitly the price of each tree.

Show that the price of *both* trees is higher when higher dividends are realized. Explain.

4. Consider a version of the growth model with **exogenously given government consumption**. There is a continuum of measure one of identical households which are endowed with an initial capital stock $k_0 > 0$ and one unit of time in each period. The utility function is given by

$$\sum_{t=0}^{\infty} \beta^t \left[\frac{c_t^{1-\sigma} - 1}{1-\sigma} + v(1 - l_t) \right],$$

where $\beta \in (0, 1)$, $\sigma > 0$, c_t is per capita consumption, l_t is per capita labor, and $v(\cdot)$ is a concave, differentiable, and increasing function.

The aggregate production function for output is given by

$$Y_t = K_t^\theta L_t^{1-\theta}.$$

There is a government that consumes an exogenously given stream $\{G_t\}_{t=0}^{\infty}$ of output and collects taxes on labor income and capital income. The tax rates are denoted by τ_t^l and τ_t^k .

- ~~a.~~ Define an Arrow–Debreu equilibrium.
- ~~b.~~ Derive the conditions that characterize an Arrow–Debreu equilibrium allocation.
- ~~c.~~ Derive the implementability constraint.
- ~~d.~~ Solving the Ramsey problem while imposing that $\tau_0^k = 0$, show that the optimal capital tax is zero from period 1 onwards: $\tau_t^k = 0 \forall t \geq 1$.